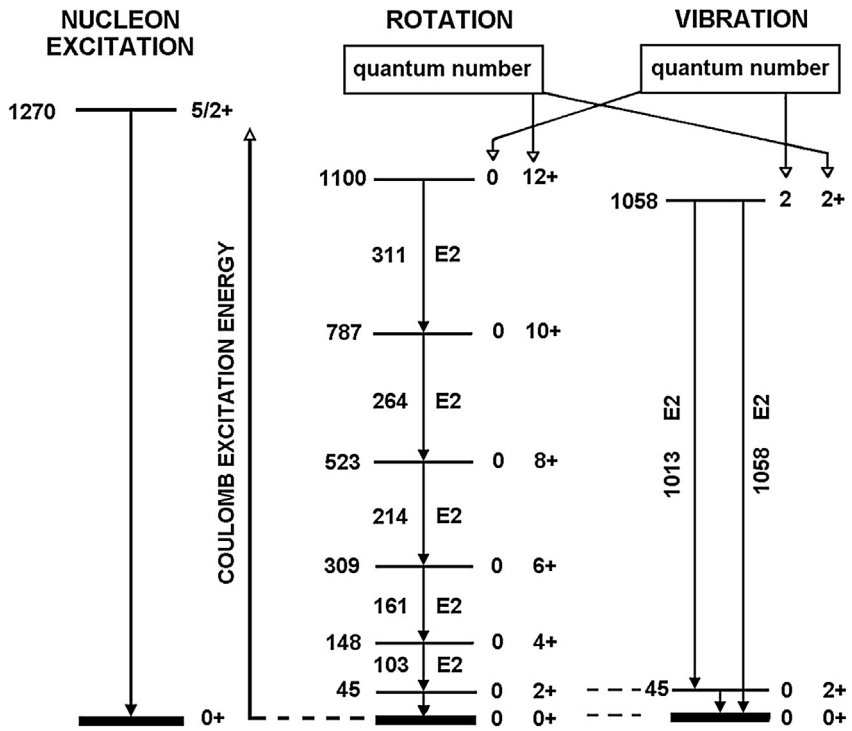


Figure 6.7 Nucleonic, rotational, and vibrational levels observed in ^{238}U; energy in keV. *(From E. K. Hyde.)*

between rotational levels involve lower energies than de-excitation from excited nuclear or vibrational states.

In case of even-even nuclei, the rotational energy levels can be often calculated from the simple expression

$$E_{\text{rot}} = \left(\hbar^2/2I_{\text{rot}}\right) n_{\text{r}}(n_{\text{r}} + 1) \tag{6.41}$$

where I_{rot} is the moment of inertia and n_{r} the rotational quantum number; this equation is identical to (3.39). The validity of this equation depends on whether the different modes of motion can be treated independently or not, which they can for strongly deformed nuclei like ^{238}U.

6.5. THE UNIFIED MODEL OF DEFORMED NUCLEI

The collective model gives a good description for even-even nuclei but cannot account for some of the discrepancy between observed spins and the spin values expected from the single-particle shell model. The latter was developed on the assumption of a nucleon

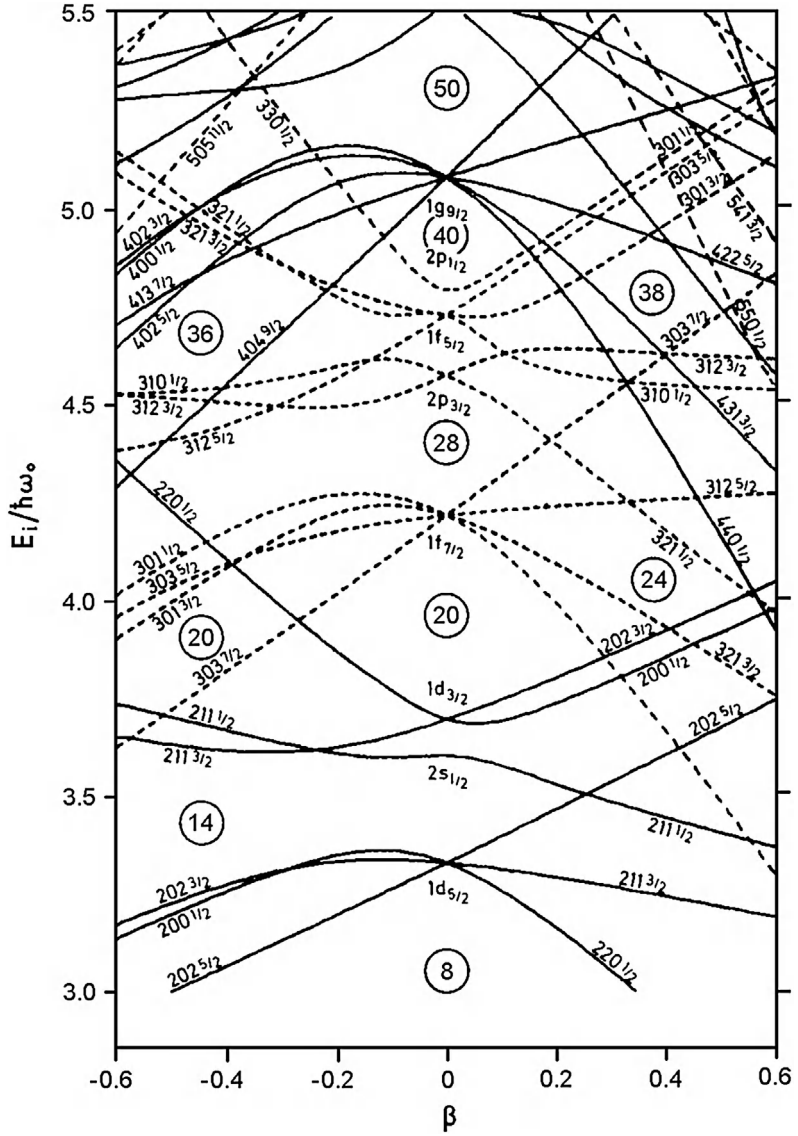


Figure 6.8 Nilsson diagram of single-particle energy levels for deformed nuclei. The energy scale is in units of $\hbar\omega_0$ ($\hbar\omega_0$ is approximately $41A^{-1/3}$ MeV). The figures along the center give the neutron or proton numbers and *l*—*j* values for the single nucleon. The figure combinations at the end of the lines are approximate quantum numbers; the first figure is the principal quantum number *n*. Full-drawn lines are for even parity, broken lines for odd. *(From Larsson, Leander, Ragnarsson and Alenius.)*

moving freely in a symmetrical potential well, a situation which is valid only for nuclei near closed shells. The angular momentum of an odd-A deformed nucleus is due both to the rotational angular momentum of the deformed core and to the angular momentum of the odd nucleon.

Consequently the energy levels for such a nucleus are different from those of the symmetric shell model.

This situation was taken into account by S. G. Nilsson, who calculated energy levels for odd nuclei as a function of the nuclear deformation β. Figure 6.8 shows how the energies of the Nilsson levels vary with the deformation β of the potential well. Each shell model level of angular momentum j splits into $j + ½$ levels (called *Nilsson levels* or *states*). Each level may contain up to two nucleons and form the ground state of a rotational band. In addition the undeformed levels ($\beta = 0$) appear in somewhat different order than for the symmetric shell model (Table 6.6). This leads to a reversal in order for some of the levels, e.g. $1f_{5/2}$ and $2p_{3/2}$ (this explains the observed level order for ^{47}Sc, Fig. 6.5). The Nilsson levels are quite different in all characteristics from the shell model states, and their prediction of energies, angular momenta, quantum numbers, and other properties agrees better with experimental data for deformed nuclei than those of any other model.

As an example we may choose $^{23}_{11}$Na, which has a quadrupole moment of 0.101 barn. Assuming the nuclear radius to be $1.1\ A^{1/3} = (a+c)/2$ (fm) we can use (6.38) and (6.39) to calculate a deformation index $\beta = 0.12$. (The value 1.1 for the constant r_o in (4.14) gives better agreement in nuclei where the inertia of the nucleus is involved.) From Figure 6.8 the 11th proton must enter the 3/2 level rather than the $1d_{5/2}$ level as the symmetric shell model indicates in Table 6.6. The experimental spin of 3/2 confirms the Nilsson prediction. Similarly, for the deformed $^{19}_{9}$F and $^{19}_{10}$Ne, we expect from Table 6.8 the odd nucleon to give spin of ½ and not 5/2, as would be obtained from Table 6.6. Again, experiment agrees with the prediction of ½.

6.6. INTERACTION BETWEEN THE NUCLEAR SPIN AND THE ELECTRON STRUCTURE

We have already seen how the spin and orbital angular momentum of the electrons and of the nucleus produce magnetic fields that interact with each other. The field produced by the electrons is much larger than that of the nucleus, and consequently the nuclear spin is oriented in relation to the field produced by the electron shell. By contrast the effects of the nuclear spin on the electron structure is so small that it usually neglected. Nuclear physics has provided us with instruments of such extreme sophistication and resolution that there are many ways of measuring with great accuracy the interaction between the nucleus and the electrons. The result has been research tools of utmost importance, most prominent being the *nuclear magnetic resonance* (nmr) techniques. The